

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 05



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Recap of Previous Lecture



Topic

Different types of keys in RDBMS

Topic

Candidate key

{ Minimal Key }

Topic

Super key =

$\left(\begin{array}{l} \text{All attributes} \\ \text{at least one} \\ \text{of C.K} \end{array} \right) + \left(\begin{array}{l} \text{0 or more} \\ \text{attributes out of} \\ \text{remaining attributes} \end{array} \right)$

Topics to be Covered



✓ **Topic** Super key

✓ **Topic** Closure of an attribute set

✓ **Topic** Candidate key identification w.r.t. given set of FDs



Q.1: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$ ✓
Find the total number of superkeys possible in relation R.

(i) when (A_1) is the only candidate key of relⁿ R

Super key = (All attributes of a candidate key) + (0 or more attributes out of remaining attributes)

= (take A_1) and { take 0 or more attributes out of remaining $(n-1)$ attributes }

= (1 way) $\times \{ {}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \dots + {}^{n-1}C_{n-1} \}$

= $(1 \times 2^{n-1})$

These are the total no. of superkeys possible

$${}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n$$

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$ ✓
Find the total number of superkeys possible in relation R.

(i) when (A_1) is the only candidate key of relⁿ R

Attributes of Relⁿ R = $A_1, A_2, A_3, A_4, \dots, A_n$
are

Possible Combinations w.r.t Super Key = $\underbrace{1 * 2 * 2 * 2 * \dots * 2}_{(n-1) \text{ times}}$

$\underbrace{1}_{A_1 \text{ must be chosen}}$
= 2^{n-1} ∴ 2^{n-1} different Superkeys are possible

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
Find the total number of superkeys possible in relation R.

(ii) when $(A_1 A_2)$ together is the only candidate key of $rel^n R$

Attributes of $Rel^n R$ are = $A_1, A_2, A_3, A_4, A_5, \dots, A_n$

Possible combinations w.r.t. Super Key =

$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	
1	1	2	2	2	$\dots$	2
Both A_1 & A_2 must be present		$(n-2)$ times				

= 2^{n-2} $\therefore 2^{n-2}$ different Super Keys are possible

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
Find the total number of superkeys possible in relation R.

(iii) When $\underline{A_1}$ and $\underline{A_2}$ are the only two candidate keys of $rel^h R$

Attributes of $rel^h R$ = $A_1 \quad A_2 \quad A_3 \quad A_4 \quad \dots \quad A_n$
are

Possible combinations
w.r.t. Super key

When A_1 is Present
but A_2 is not present

When A_1 is not present
but A_2 is present

When both A_1 & A_2
are Present

$$\begin{aligned}
 & \text{(or)} \quad \begin{pmatrix} \checkmark & \times \\ 1 & 1 \end{pmatrix} * 2 * 2 * \dots * 2 = 2^{n-2} \\
 & \text{(or)} \quad \begin{pmatrix} \times & \checkmark \\ 1 & 1 \end{pmatrix} * 2 * 2 * \dots * 2 = 2^{n-2} \\
 & \text{(or)} \quad \begin{pmatrix} \checkmark & \checkmark \\ 1 & 1 \end{pmatrix} * 2 * 2 * \dots * 2 = 2^{n-2} \\
 & \hline
 & 3 \cdot 2^{n-2}
 \end{aligned}$$

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
Find the total number of superkeys possible in relation R.

(iii) When (A_1) and (A_2) are the only two Candidate keys of relⁿ R
Total no. of Superkeys in which A_1 is present



2^{n-2} This area represent the number of Superkeys in which both A_1 & A_2 are present

$$\begin{aligned}\text{Total No. of S.K.} &= 2^{n-1} + 2^{n-1} - 2^{n-2} \\ &= 2 \cdot 2^{n-2} + 2 \cdot 2^{n-2} - 2^{n-2} \\ &= 4 \cdot 2^{n-2} - 2^{n-2} \\ &= 3 \cdot 2^{n-2}\end{aligned}$$

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
 Find the total number of superkeys possible in relation R.
 (iv) When (A_1A_2) & (A_2A_3) are the only two candidate keys of relⁿ R

Attributes of Relⁿ R = $A_1 \quad A_2 \quad A_3 \quad A_4 \quad A_5 \dots A_n$
 are

Possible Combinations w.r.t. SuperKey =

(or)	$\checkmark(1)$	$\checkmark(1)$	$\times(1)$	$\times 2 \times 2 \times \dots \times 2$	$= 2^{n-3}$
(or)	$\times(1)$	$\checkmark(1)$	$\checkmark(1)$	$\times 2 \times 2 \times \dots \times 2$	$= 2^{n-3}$
(or)	$\checkmark(1)$	$\checkmark(1)$	$\checkmark(1)$	$\times 2 \times 2 \times \dots \times 2$	$= 2^{n-3}$

$\underbrace{\hspace{10em}}_{n-3 \text{ times}}$
 $3 \cdot 2^{n-3}$
 Total no. of Superkeys possible

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
 Find the total number of superkeys possible in relation R.
 (iv) When (A_1A_2) & (A_2A_3) are the only two candidate keys of relⁿ R



$$\begin{aligned}
 \text{Total no. of Superkeys} &= 2^{n-2} + 2^{n-2} - 2^{n-3} \\
 &= 2 \cdot 2^{n-3} + 2 \cdot 2^{n-3} - 2^{n-3} \\
 &= 4 \cdot 2^{n-3} - 2^{n-3} \\
 &= 3 \cdot 2^{n-3}
 \end{aligned}$$

HW Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
Find the total number of superkeys possible in relation R.

(V) When (A_1A_2) & (A_3A_4) are the only two candidate keys of $rel^h R$

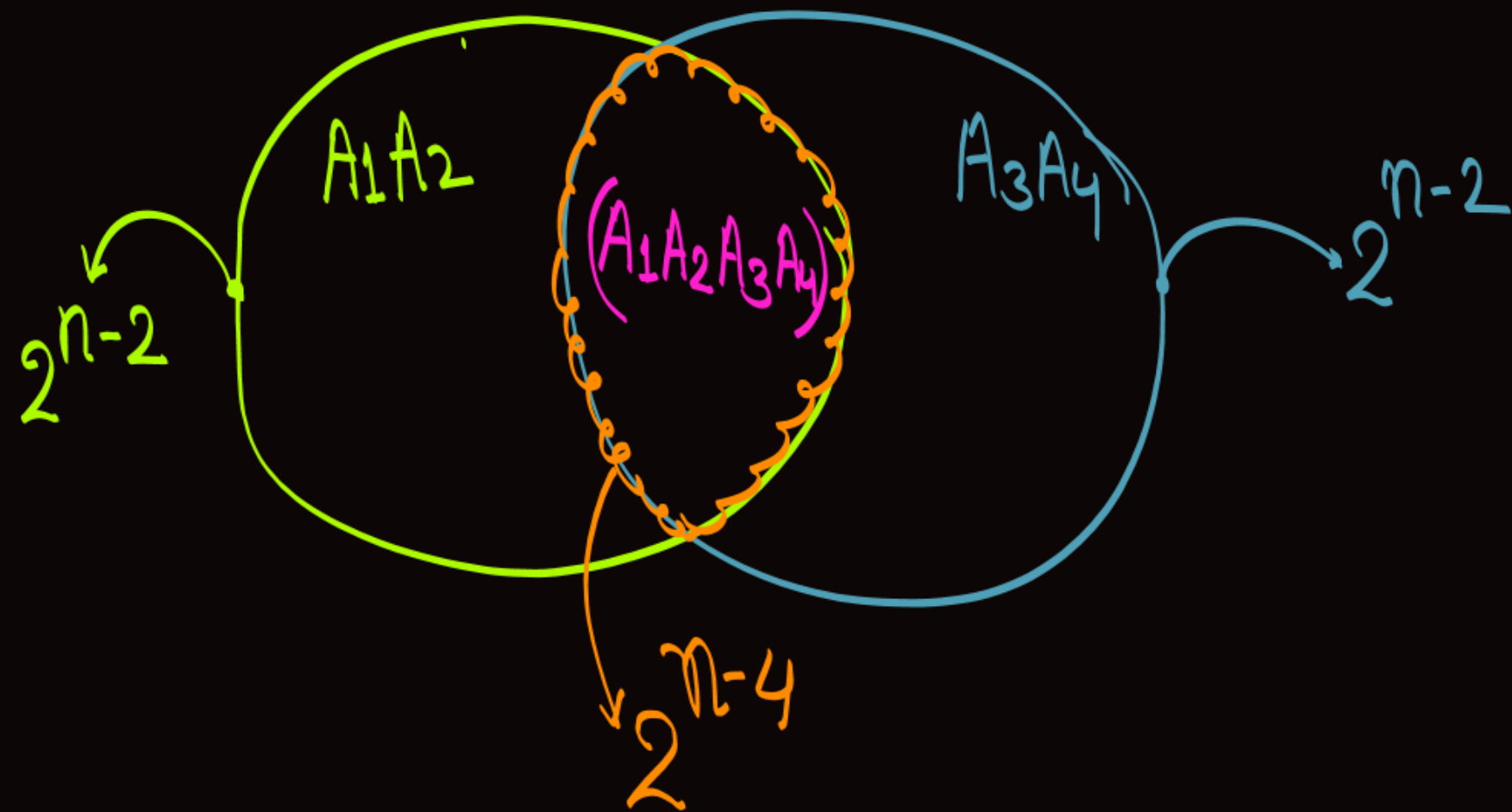
Attributes of $Rel^h R$ are = $A_1 \quad A_2 \quad A_3 \quad A_4 \quad A_5 \quad A_6 \quad \dots \quad A_n$
(n-4) times

Possible combinations w.r.t. Super Key =

(or)	✓	✓	x	x	* 2	* 2	* - - 2	= 2^{n-4}
(or)	✓	✓	✓	x	* 2	* 2	* - - 2	= 2^{n-4}
(or)	✓	✓	x	✓	* 2	* 2	* - - 2	= 2^{n-4}
(or)	✓	✓	x	✓	* 2	* 2	* - - 2	= 2^{n-4}
(or)	✓	✓	✓	✓	* 2	* 2	* - - 2	= 2^{n-4}
(or)	x	x	✓	✓	* 2	* 2	* - - 2	= 2^{n-4}
(or)	✓	x	✓	✓	* 2	* 2	* - - * 2	= 2^{n-4}
(or)	x	✓	✓	✓	* 2	* 2	* - - * 2	= 2^{n-4}

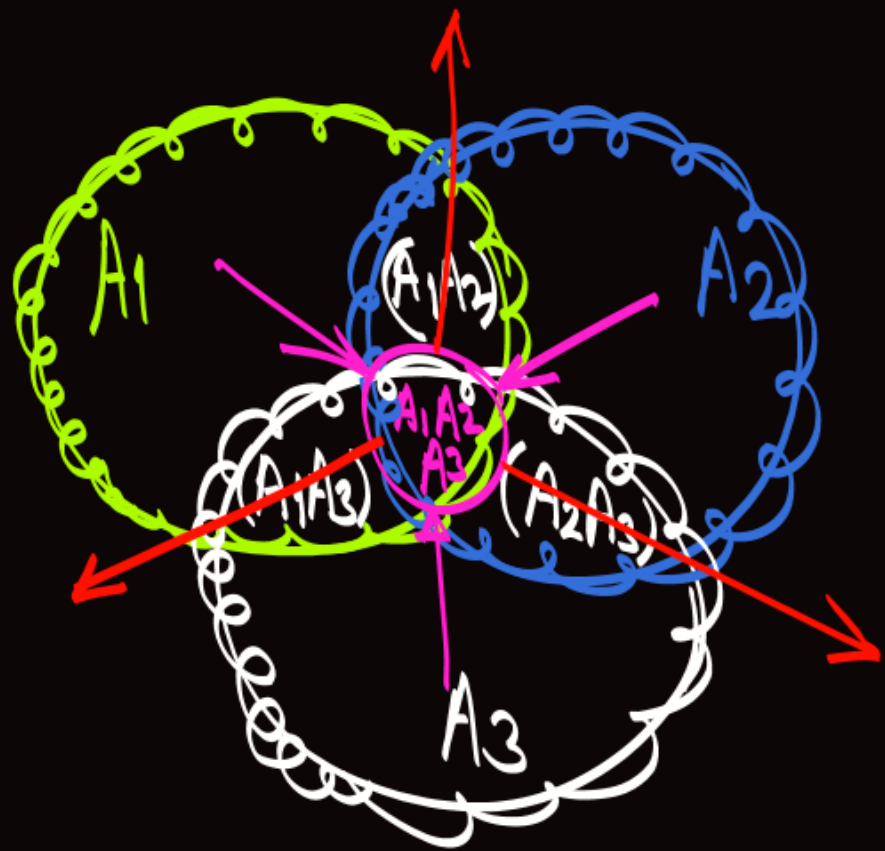
Total no. of Super Keys = $7 \cdot 2^{n-4}$

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
 Find the total number of superkeys possible in relation R.
 (V) When (A_1A_2) & (A_3A_4) are the only two candidate keys of relⁿ R



$$\begin{aligned}
 \text{Total no. of Super keys} &= 2^{n-2} + 2^{n-2} - 2^{n-4} \\
 &= 4 \cdot 2^{n-4} + 4 \cdot 2^{n-4} - 2^{n-4} \\
 &= 8 \cdot 2^{n-4} - 2^{n-4} \\
 &= \underline{\underline{7 \cdot 2^{n-4}}} \quad \underline{\underline{Ans}}
 \end{aligned}$$

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_n)$
 Find the total number of superkeys possible in relation R.
 (vi) When $(A_1), (A_2)$ & (A_3) are the three candidate keys of relⁿ R



Total no. of Superkeys

$$= 2^{n-1} + 2^{n-1} + 2^{n-1} - 2^{n-2} - 2^{n-2} - 2^{n-2} + 2^{n-3} = 7 \cdot 2^{n-3}$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_N)$
Find the total number of superkeys possible in relation R.
When each attribute of relation is a Candidate key.

→ When each attributes of a relation is a Candidate key
then Every non-empty subset of attributes will be a
Super Key.

as Total no. of Superkeys = $2^n - 1$

$\nwarrow$ Total no. of subsets n

$\nwarrow$ Subtract Empty subset

$= 2^n - 1$



Topic : Closure of an attribute set

Closure of an attribute set X (i.e., X^+) can be defined as set of all the attributes which can be functionally determined from attributes of set X .

#e.g. Assume a relation R (A,B,C,D) that has the following functional

dependencies:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

$$(A)^+ = \{A, B, C, D\}$$

$$(AB)^+ = \{A, B, C, D\}$$

$$(AC)^+ = \{A, B, C, D\}$$

$$(AD)^+ = \{A, B, C, D\}$$

$$(ABC)^+ = \{A, B, C, D\}$$

$$(ABD)^+ = \{A, B, C, D\}$$

$$(ACD)^+ = \{A, B, C, D\}$$

$$(B)^+ = \{B, C, D\}$$

$$(BC)^+ = \{B, C, D\}$$

$$(BD)^+ = \{B, C, D\}$$

$$(BCD)^+ = \{B, C, D\}$$

$$(C)^+ = \{C, D\}$$

$$(CD)^+ = \{C, D\}$$

$$(D)^+ = \{D\}$$



Topic : Super key



Let R be the relational schema, and let X be some attribute set over relation R . If X^+ Contains all attributes of relation R , then X is called super key of relation R .

#e.g. Assume a relation R (A,B,C,D) that has the following functional

dependencies:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

$$(A)^+ = \{A, B, C, D\}$$

All attributes of R

$\therefore$ A is a Super Key of R

$$(AB)^+ = \{A, B, C, D\}$$

$\therefore$ AB is also a Superkey of R

$$(AC)^+ = \text{All attributes of R}$$

$$AC \text{ --- } \underline{\hspace{2cm}}$$

$$(AD)^+ = \text{--- } \underline{\hspace{2cm}}$$

$$\therefore AD \text{ --- } \underline{\hspace{2cm}}$$

$$(ABC)^+ = \text{--- } \underline{\hspace{2cm}}$$

$$\therefore ABC \text{ --- } \underline{\hspace{2cm}}$$

$$(ABD)^+ = \text{--- } \underline{\hspace{2cm}}$$

$$\therefore ABD \text{ --- } \underline{\hspace{2cm}}$$

$$(ACD)^+ = \text{--- } \underline{\hspace{2cm}}$$

$$\therefore ACD \text{ --- } \underline{\hspace{2cm}}$$

$$(ABCD)^+ = \{A, B, C, D\}$$

$$\therefore ABCD \text{ --- } \underline{\hspace{2cm}}$$

• Let "X" be some set of attributes from relation R.

• if $(X)^+ =$ All attributes of relation R, then X is Superkey of R.

• If X is a Superkey of R and no proper subset of X is a Superkey of relation R then we can say that

X is a minimal Super key

i.e. X is a Candidate Key of relation R.



Topic : Candidate key (Minimal Super key)

- Let R be the relational schema, and let X be the super key of relation R .
If no proper subset of X is a super key then X is minimal super key
i.e., X is Candidate key

eg. let $(AB)^+ =$ All attributes of relation R , so (AB) is a Superkey of R

Non-Empty proper
Subsets of AB are $\left. \begin{array}{l} \{A\}, \text{ if } (A)^+ = \text{Not all attributes of } R \\ \text{and} \\ \{B\}, \text{ if } (B)^+ = \text{Not all attributes of } R \end{array} \right\} \Rightarrow$ i.e. No proper subset of (AB) is a S.K.
Hence, (AB) will be a Candidate Key

#Q. Assume a relation $R(A, B, C, D, E)$ that has the following functional dependencies:

$AB \rightarrow C,$

$B \rightarrow E,$

$C \rightarrow D$

Here A & B are essential attributes of the relation

Find the Candidate key of R .

The attributes of the relation which are not present in the R.H.S. Part of any FD are called "Essential attributes" of the relation

Note: Every essential attribute must be present in Every Candidate key of the relation

#Q. Assume a relation R (A, B, C, D, E) that has the following functional

dependencies:

AB → C,

B → E,

C → D

Find the Candidate key of R.

$(AB)^+ = \{A, B, C, D, E\}$ ∴ AB is a S.K.
all attributes

* Check for C.K.

Proper Subsets of (AB) → $(A)^+ = \{A\}$ Not all attributes ∴ "A" is not a S.K.
→ $(B)^+ = \{B, E\}$ Not all attributes ∴ B is not a S.K.

$(AB)^+ = \{A, B, C, E, D\}$

all attributes of Relation

∴ AB is a Super key of relation
{ we know Both A & B are Essential
∴ No need to check w.r.t.
Proper subsets of AB.

Ans

AB is a Candidate Key.

None of the proper subset of (AB) is a S.K.
∴ AB is a Candidate Key

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

$AB \rightarrow C$,

$B \rightarrow E$,

$C \rightarrow D$

Find the Candidate key of R.

$\boxed{AB}$ is a Candidate Key of the relation

$\therefore$ Prime attributes = $\{A, B\}$

observe that, None of the prime attribute is present in R.H.S. part of any FD of the given FD set

Hence $\boxed{AB}$ will be the only Candidate Key of given Relⁿ.

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

$AB \rightarrow C,$
 $B \rightarrow E,$
 $C \rightarrow D$
 $E \rightarrow A$

"B" is not present in RHS. part of any FD
 ∴ "B" is an Essential attribute.

$(B)^+ = \{ B, E, A, C, D \}$

all attributes

Find the Candidate key of R.

Prime Attribute = $\{ B \}$

Not present in RHS of any FD

∴ $\boxed{B}$ is the only C.K.

∴ B is a Super Key.

"B" is a key with single attribute
 ∴ $\boxed{B}$ is a Candidate Key

#Q. Assume a relation R (A, B, C, D, E) that has the following functional

dependencies: $\rightarrow (AB)^+ = \{A, B, C, D, E\}$ $\therefore AB$ is a Super Key
all attributes

$AB \rightarrow C,$

$B \rightarrow E,$

$C \rightarrow D$

$E \rightarrow A$

• Check for Candidate Key

Proper subsets
of AB

$(A)^+ = \{A\}$

Not all attributes $\therefore$ Not a SK

$(B)^+ = \{B, E, A, C, D\}$

All attributes $\therefore$

B is a SK

One of the proper
subset of AB is a SK

$\therefore AB$ is just a SK,
it is not a CK

Find the Candidate key of R.

B is a Super Key

$\rightarrow$ Now check if B is a C.K. or not

"B" is a key with a single attribute

$\therefore \boxed{B}$ is a Candidate Key

Prime Attribute = $\{B\}$



2 mins Summary



Topic

Super key ✓

Topic

Closure of an attribute set ✓

Topic

Candidate key identification w.r.t. given set of FDs ✓



THANK - YOU